EXP 12(c) CHAT CLIENT-SERVER USING TCP/IP. SOCKETS.

24-09-26

AIM:-

To implement a chat client-server application using socket and enable communication between multiple clients through a server.

PROCEDURE:-

1. Create a chat server program using Socket programming.
2. Start and run the chat server.
3. Create the chat client program and connect it to the server.
4. Run two or more client program in separate terminals
5. Enter a name for each client.
6. Send message from one client to the server.
7. The server broadcasts the message to the other connected clients.
8. Verify that the clients are exchanging message through the server.
9. Continue the communication between the clients.
10. Enter/quit to terminate a client connection.

RESULT:-

Thus the chat client-server application was successfully implemented and message was successfully exchanged blw multiple clients through the server.

EXP 12C

CODE:-

```
import java.io;
import java.net.*;
public class ChatClient{
    public static void main(String args[])
        Scanner sc = new Scanner(System.in);
    String host = "Localhost";
    int port = 5000;
    Socket socket = new Socket(host,port);
    BufferedReader in = new BufferedReader(
            socket.getInputStream()));
    PrintWriter out = new PrintWriter(
            new OutputStreamWriter(socket.getOutput
                                   Stream()),
    try{
      String message;
      while(message = sc.nextline() != null){
          System.out.println(message);
      if(message.equalsIgnoreCase("/quit"))
        break;
      }
      socket.close()
      System.out.println("Chat Client closed");
}
```

```
PS C:\Users\Luqman\Downloads\CN_EXPT_12> java ChatServer
Chat server started on port 5000
Mr. X connected from /127.0.0.1
Mr. Y connected from /127.0.0.1
Mr: X: Hello, How are you?
Mr. Y: I am Fine
```

Problems Output Debug Console Terminal Ports Jupyter

```
PS C:\Users\Luqman\Downloads\CN_EXPT_12> java ChatClient
Enter your name:
Mr. X
Welcome, Mr. X! Type /quit to exit.
Mr. Y has joined the chat.
Hello, How are you?
Mr. Y: I am Fine
```

```
PS C:\Users\Luqman\Downloads\CN_EXPT_12> java ChatClient
Enter your name:
Mr. X has joined the chat.
Mr. Y
Welcome, Mr. Y! Type /quit to exit.
Mr. X: Hello, How are you?
I am Fine
```

Scanned with OKEN Scanner